

Microbiology Galaxy Lab (MGL): The first community-driven gateway for reproducible and FAIR analysis of microbiological data

Engy Nasr¹, Nikos Pechlivanis², Nikolaos Strepis³, microGalaxy Community, Paul Zierep¹ and Bérénice Batut⁴

¹Bioinformatics Group, Department of Computer Science, University of Freiburg, Georges-Koehler-Allee 106, D-79110 Freiburg, Germany, ²Institute of Applied Biosciences, Centre for Research and Technology Hellas, Thessaloniki, Greece, ³Department of Pathology and Clinical Bioinformatics, Erasmus MC Cancer Institute, Erasmus MC, Rotterdam, Netherlands, ⁴(Aero)microbiome bioinformatics researcher, Clermont-Ferrand, Auvergne-Rhône-Alpes, France.

Introduction
Microbial research increasingly relies on high-throughput and multi-omics approaches, generating complex datasets. However, identifying and navigating relevant tools and workflows within platforms like Galaxy remains challenging.

How do scientists analyse microbial data?

130 responses of the survey conducted among the microbiology research community (March - September 2023)

What is your main target?

Bacteria	44.35%
Eukaryotes	22.58%
Mixed Communities	19.35%
Other	7.26%
Virus	6.45%

Which techniques do you use?

Single Organism Genomics	23.89%
Metagenomics	23.08%
(Meta)Transcriptomics	22.27%
Metabarcoding	14.98%
(Meta)Proteomics	8.91%
Metabolomics	3.64%
Other	3.24%

What analysis do you use or would like to do?

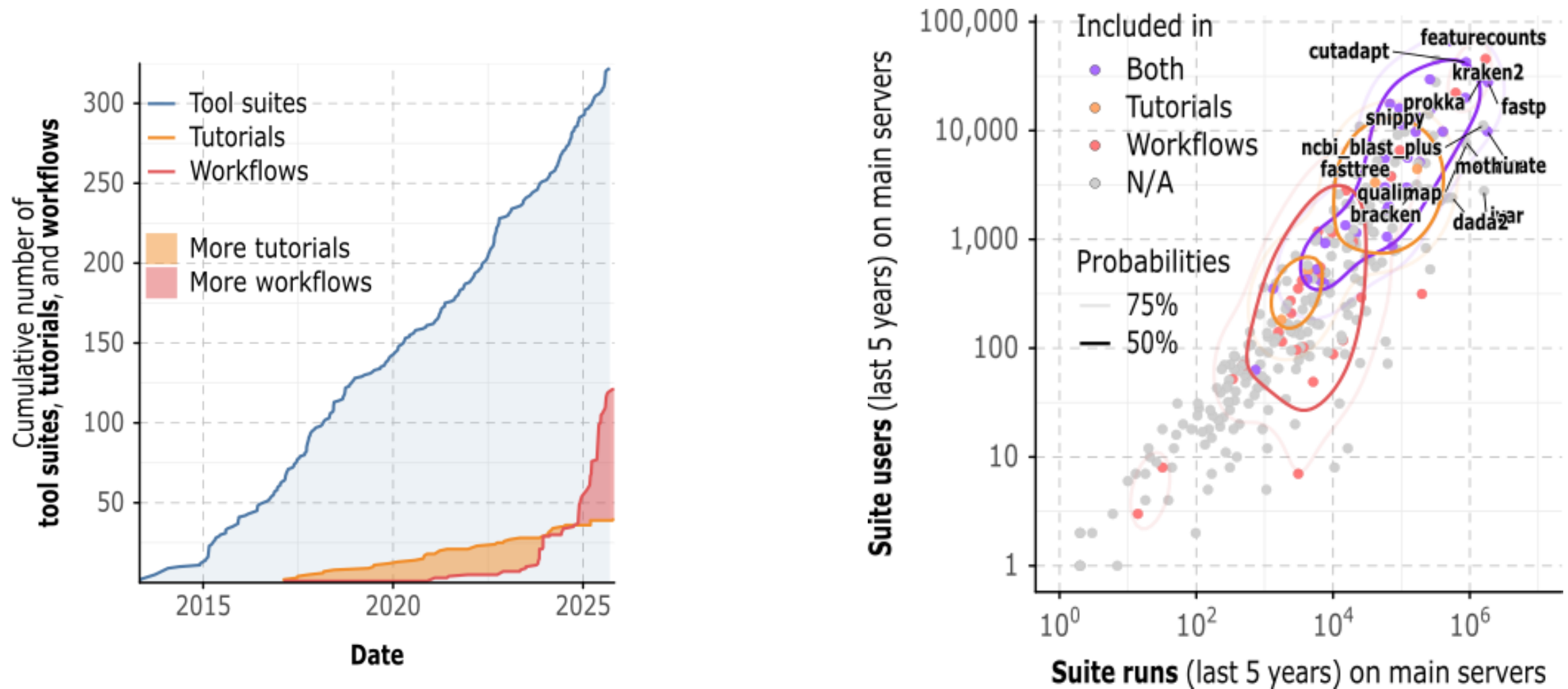
Yes	77.69%
No	11.54%
Unknown	10.77%

Number of Responses

Scientists study different organisms using different techniques and analyses

Could scientists use Galaxy for microbial data analysis?

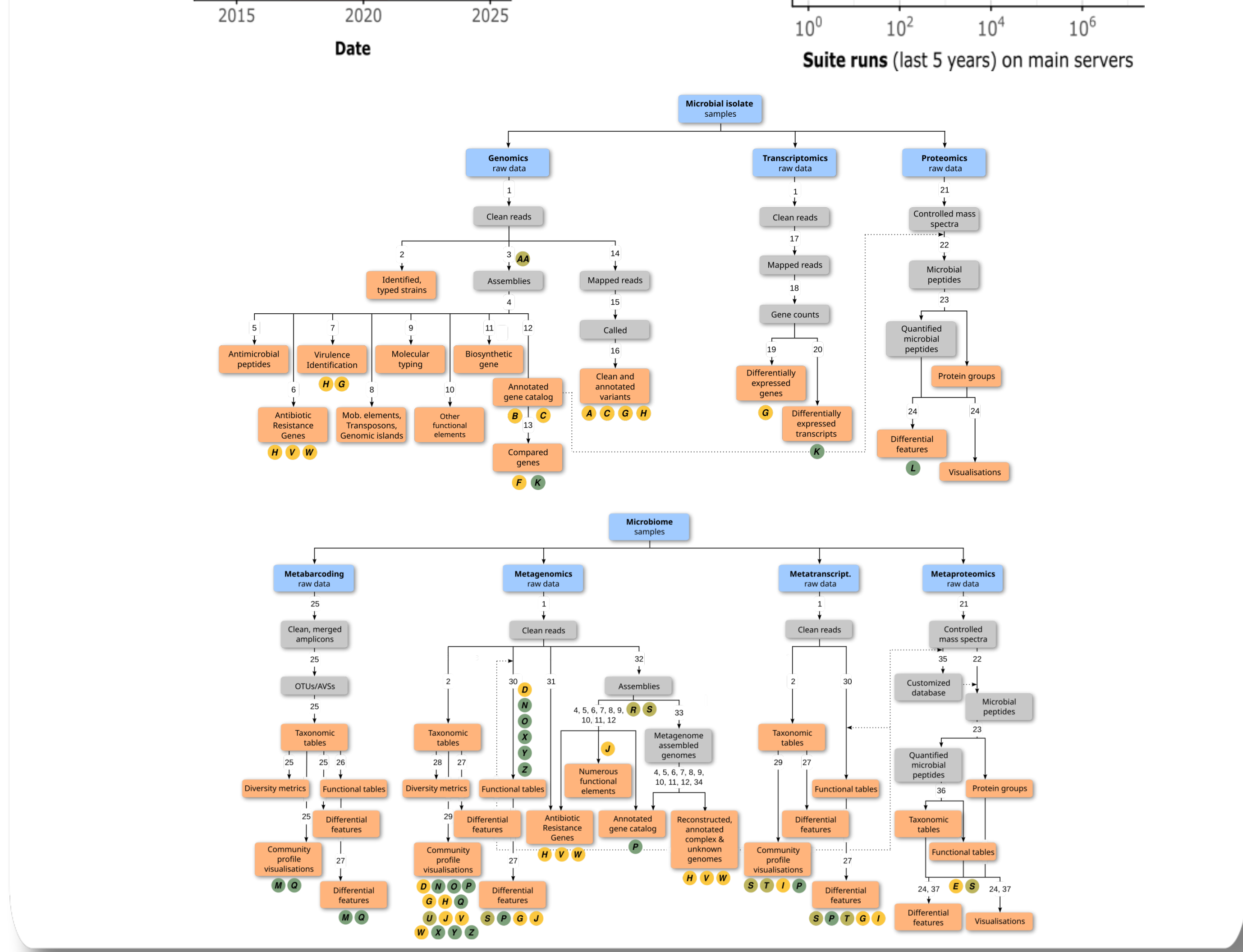
Resources for microbial analysis in Galaxy are highly used for a wide range of key computational functions



Do scientists use Galaxy for publications?

28% of publications citing Galaxy are microbiology-related papers

Scientists study different organisms using different techniques and analyses and want to use Galaxy



How does MGL help scientists find the right resources for microbial data analysis?

A Tools sorted by type of analysis and/or tool suites

Direct links to resources (tools, workflows, training material) for **Microbial Isolates** data analysis

Direct links to resources (tools, workflows, training material) for **Microbiome** data analysis

Links to curated tools grouped by EDAM operations

B History of analysis and files

Links to curated workflows grouped by FAIRness levels

Links to support and help

Links to the community

C Contributors

MGL is the first community-driven gateway for reproducible and FAIR analysis of microbiological data

MGL has:

- 884 microbiology related tools
- 117 workflows
- 39 tutorials

MGL is FAIR:

- Findable (e.g., WorkflowHub, Dockstore)
- Accessible (Galaxy servers: org, .eu, .org.au, .fr)
- Interoperable (EDAM)
- Reusable (e.g., versioned workflows)

MGL sustainability:

- MGL is a microGalaxy Community-driven ecosystem
- MGL Roadmap 2026–2030

MGL availability:

microbiology.usegalaxy.org, .eu, .org.au, .fr (and more Galaxy servers to come)

For more information about the MGL and our roadmap, check our preprint